

AIMS System - Detailed Class Design Documentation

16. Design Class: Invoice

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	invoiceId	int	null	Unique identifier for the invoice
2	order	Order	null	Order associated with this invoice
3	issueDate	Date	current date	Date when invoice was created
4	productList	List<OrderItem>	empty list	List of products in the invoice
5	subtotal	float	0.0	Total price excluding VAT
6	VAT	float	0.0	Value Added Tax amount (10% of subtotal)
7	totalWithVAT	float	0.0	Subtotal + VAT
8	regularDeliveryFee	float	0.0	Fee for regular delivery items
9	rushDeliveryFee	float	0.0	Fee for rush delivery items
10	totalAmount	float	0.0	Final amount to be paid
11	paymentMethod	String	"Credit Card"	Method of payment
12	recipientInfo	DeliveryInfo	null	Delivery information of the recipient
13	VAT_RATE	float	0.1	Value-added tax rate (constant 10%)

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	createInvoice	Invoice	Creates a new invoice with order and delivery information
2	calculateSubtotal	float	Calculates the subtotal price of all products
3	calculateVAT	float	Calculates the VAT amount based on subtotal
4	calculateTotalWithVAT	float	Calculates the total amount including VAT
5	calculateDeliveryFees	void	Calculates separate regular and rush delivery fees
6	calculateTotalAmount	float	Calculates the final total amount to be paid
7	generateInvoiceDetails	Map<String, Object>	Generates a complete map of invoice details
8	save	void	Saves the invoice to the system
9	getInvoiceById	Invoice	Retrieves an invoice by its ID
10	Invoice	void	Constructor for creating an invoice

Parameter:

- **createInvoice(cart: Cart) : Invoice**
 - cart: The user’s cart data.
 - Return: A fully built Invoice with items, total amounts, and an assigned ID.

Exception:

- InvoiceSaveException – Thrown if saving the invoice fails (database error, disk error, etc.).

- InvalidCartException – If cart is empty or invalid when calling createInvoice().

Method:

- Invoice() (constructor)
 - Purpose: Sets up default values (e.g., status = "unpaid").
 - Usage: Called internally or externally to create an invoice instance.
- createInvoice(cart: Cart): Invoice
 - Purpose: Takes a cart object, calculates subtotal, VAT, and total, sets invoice items, then returns an Invoice.
 - Usage: Typically called by InvoiceScreen after the user finalizes the cart.
- save()
 - Purpose: Persists the invoice data (items, amounts, status) into a database or storage.
 - Usage: Called once the invoice is ready to be stored or updated (e.g., after changing status).

State

- status transitions: “unpaid” → “paid” → possibly “cancelled.”
- $\text{totalAmountInclVAT} = \text{totalAmountExclVAT} * 1.1$ (assuming 10% VAT).
- $\text{finalAmount} = \text{totalAmountInclVAT} + \text{deliveryFee}$.